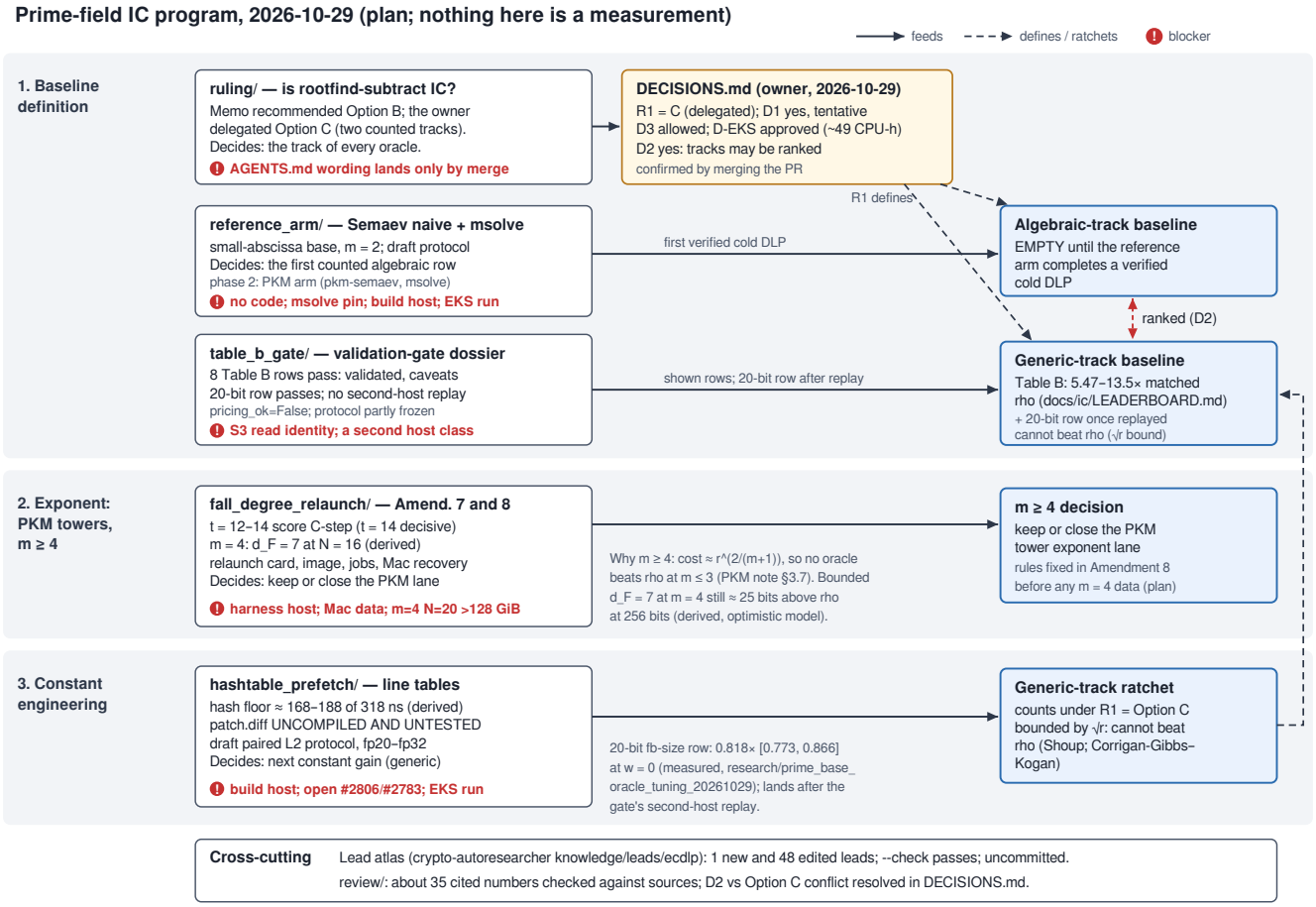
Prime-field index calculus: the program as of 2026-10-29

This directory is preparation, not measurement. Nothing in it was run as an experiment. It reports no new result, and it changes no row of the IC leaderboard (`docs/ic/LEADERBOARD.md`, `docs/ic/leaderboard.json`, `docs/ic-leaderboard.html`), of `benchmarks/README.md`, or of any `VALIDATION.json`. Every number below is one of three kinds, and is labelled: **measured** (quoted from a committed file, whose repo-relative path is given), **derived** (worked out by hand or by a track from quoted values), or **plan**. Checkout: crypto `main` at `7ffaab06b`.

The program has six prep tracks. Five sit in subdirectories here. The sixth is an edit of the ECDLP lead atlas in the sibling checkout. A cross-track review (`review/`) checked them afterwards, and the owner's answers so far are in `DECISIONS.md`. The tracks fall into three lanes:

1. **Baseline definition.** What counts as prime-field IC, which baseline each result ratchets, and whether the existing rows pass the validation gate. Tracks: `ruling/`, `reference_arm/`, `table_b_gate/`.
2. **Exponent, through PKM towers at m ≥ 4.** The only route on record that could change the exponent of prime-field IC rather than its constant. Track: `fall_degree_relaunch/` (and phase 2 of `reference_arm/`).
3. **Constant engineering.** Making the existing relation trial cheaper. Track: `hashtable_prefetch/` (and the 20-bit factor-base row that `table_b_gate/` takes to its second-host replay).



`program.svg` shows the tracks as boxes, what each one feeds as arrows, and each blocker as a red mark. Solid arrows mean "produces input for"; dashed arrows mean "defines" or "ratchets".

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1. Where things stand

Incumbent prime rows (measured, `docs/ic/LEADERBOARD.md` Table B). The eight prime-field rows, `ic1-fp10` to `ic1-fp24` ($\log_2 r$ = 9.7 to 23.4), are 5.47× to 13.5× matched rho with the negation map. All use m = 2 and the `mitm_m2_negfold_walk*` pair-table oracle (`docs/ic/leaderboard.json`).

Newest constant gain (measured, `research/prime_base_oracle_tuning_20261029/README.md`). At 20 bits, growing the factor base from N = 32 to N = 45 gives an S ratio of 0.818 [0.773, 0.866] at w = 0 and 0.726 [0.677, 0.778] at w = 1. The gap to rho narrows from 9.47× to 7.75× [6.92, 8.70]. The 24- and 28-bit selections are relabelling, not improvement. The row is in `benchmarks/20261029-prime-fb-size-20bit` but not on the leaderboard: its

protocol (§13) asks for a second-host replay first.

Owner decisions taken today (`DECISIONS.md`).

#	question	decision
R1	Is `rootfind-subtract` index calculus?	**Option C, two counted tracks** (delegated; confirmed by merging the PR that carries `DECISIONS.md`). The generic track holds table, pair-table and root-finding decomposition: Table B, `rootfind-*`, `mitm`, `subtract`, the 20-bit gain. Its gains count and ratchet, labelled "generic: bounded by √r; cannot beat rho". The algebraic track holds relations from a solver that receives the factor-base constraint inside the system. Its prime-field baseline is **empty** until the reference arm runs.
D1	Is the small-abscissa base, with F_V = ∏(x – c), "algebraically defined"?	**Yes, tentatively.** So the Semaev-naive/msolve arm is the counted algebraic reference.
D2	May the Table B incumbent be ranked against the algebraic reference?	**Yes** (owner). R1 is amended to match: the two tracks may be ranked against each other on the same instances, in the same unit and across the §5a band, and each track ratchets only its own baseline (`DECISIONS.md`). The reference arm's rule C3 changes from "context only" to a ranked comparison at the freeze.
D3	Pinned, vendored msolve counter patch	**Allowed** (delegated); its sha256 joins the candidate identity.
D-EKS	About 49 CPU-hours (derived; range 15–150) on EKS Spot for the reference arm	**Approved.** This covers the reference arm only.

Until that PR merges, `AGENTS.md` and `docs/ic/SWEEPER.md` still carry the 2026-10-24 wording: `prime-table` is reference only and `prime-rootfind` is unclassified. The tracks were written against that wording. The review then added notes pointing to `DECISIONS.md`.

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2. Lane 1: baseline definition

2.1 `ruling/`: does the root-finding oracle count as IC?

Delivers. A decision memo (`ruling/README.md`) and a machine-readable classification of every prime oracle under three candidate wordings (`ruling/classification.json`, with `adopted_option` now set to C).

What it found. At m = 2, `rootfind-subtract` does one batched inversion per target. It computes x(R ∓ F_i) for every base abscissa, which are exactly the two roots of $S_3(x_R, x_i, X)$, and looks each one up in a hash index of the base (code reading: `ic_framework/plugins.rs` lines 2238–2521). Its cost is the subtract oracle's. In the 2026-10-29 train cells `rfsub2` and `sub2` have S = 579.571 and 579.547 at 20 bits (measured, `research/prime_base_oracle_tuning_20261029/select/band-train-P20.md`), and the oracle would run the same on a random base. The memo argues both sides as strongly as it can. It recommended Option B (root finding is a table method, not IC). The owner chose Option C instead, which counts root finding on a separate generic track.

Decides. The track of every oracle (§5.4 table, column C), and therefore which baseline the gate work and the prefetch work ratchet: the generic one. It does not change the eight Table B rows, which were never root-finding rows.

Blockers. The wording reaches `AGENTS.md` only by merge (the owner, by merging). The follow-up code relabel needs a Conductor task: `Track::Generic` in `src/bin/ic/sweep/space.rs`, the `docs/ic/SWEEPER.md` table and the board labels in `scripts/build_ic_leaderboard.py`. That script's line 2380 still says prime curves have no algebraic oracle, though `pkm-semaev` is wired at `ecbench/methods.rs` ~4171. The D2 conflict is resolved (O1; `DECISIONS.md`).

2.2 `reference_arm/`: the published algebraic reference

Delivers. A recommendation (`README.md`), a draft protocol (`PROTOCOL.md`; `protocol.draft.json` passes `tools/experiment_protocol.py validate` as a draft) and an implementation spec (`IMPLEMENTATION.md`).

What it proposes (plan). Use Semaev's naive prime-field IC as the reference:

- the small-abscissa base `prime-abscissa:size=N`, the same base the incumbent uses;
- m = 2, and m = 3 at 10–18 bits;
- each trial solves ⟨S_{m+1}(x_1…x_m, x_R), F_V(x_i)⟩ with `msolve -P 1` over F_p;
- each trial is priced as cold whole-solve S in GAE, with msolve's multiply-adds placed in the §5a band.

Because base and trial stream are shared, reference and incumbent differ only in the decomposition oracle. That gives exact checks: identical hit sets, and byte-identical relation streams against a grid engine. The arms are rho-neg (+A/A), `inc`, `pub-m2`, `pub-m2-tuned`, a planted positive control, a grid negative control, an `f4_fp` engine arm and m = 3 arms. The panel is the Table B roster at 10–24 bits plus today's 20- and 24-bit holdout curves. PKM's tower base

(`pkm-semaev` on `prime-kummer` with an msolve engine) is phase 2. FGHR 2-torsion symmetrisation does not apply: the Table B curves have odd order (`docs/curves/registry.json`, checked by the review).

Decides. The first counted row of the algebraic track, once it completes a verified cold DLP.

Blockers.

- **No code exists.** It needs an F_p solver plug point (`fp_solvers.rs`), `msolve.rs`, `semaev_fp.rs` and `semaev_naive.rs`, plus edits across `ecbench`. Unblocked by an implementation session on a host with a Rust toolchain and several GB of disk.
- **No sound msolve build.** No msolve build is known to be sound above $p = 2^{16}$: Ubuntu's 0.6.5-1build2 is documented as wrong there (crypto-autoresearcher `msolve.py`). Unblocked by a pinned, static, counter-patched build (candidate commit `a18d0de4`) with a self-test.
- **The images lack msolve.** The ecbench EKS image and the CI replay workflows would need msolve added. Unblocked by an implementation PR merged to `main`.
- **Budget not yet measured.** The budget is about 49 CPU-hours, range 15–150 (derived), and RAM is not measured. A preflight replaces both.

2.3 `table_b_gate/`: do the existing rows pass the validation gate?

Delivers. A dossier (`README.md`), a recomputation script and its output (`gate_status.py`, `gate_status.out.txt`), and a band figure (`band_ratio.svg`, `plot_band.py`).

What it found (measured, from the committed evidence).

- **Table B passes the gate.** All eight Table B prime rows pass: `docs/ic/runs/VALIDATION.json` is complete, and `validation_gate.py evidence docs/ic/runs` prints `valid`.
- **Two caveats remain, display only.** `pricing_ok=False`: IC's native work is priced by a calibration derived from wall time, while rho's canonicalisations are uncharged. `protocol_frozen_ok=partial`: the recipe is the minimum of 9 on the same 3 targets, and Round 5 was never frozen.
- **The order holds at every weight.** Across the band, IC/rho runs from 4.66 to 16.30, and IC > rho at every weight.
- **The 20-bit row passes too, but is not on the leaderboard.** It passes the gate mechanically and is in `benchmarks/README.md`. Its replay certificate is same-host (672/672 identical, `same_host_class_as_measurement: true`).

Decides. Which rows the generic-track baseline may show, and the exact steps for the second-host replay that admits the 20-bit row:

1. Hydrate `records.jsonl` from S3.
2. Build ecbench at `0e124fb1`.
3. On a host class other than `ECBENV3hc4d998f5ae3f`, run `ecbench verify --replay-all`.
4. Accept only 672/672 identical.

Blockers.

- **S3 read.** Hydrating needs S3 read on the artifact store. Unblocked by an executor with the lab AWS identity.
- **A second host class.** The custom-command route `isolab_k8s_run` can run the replay; the ecbench-only request routes cannot. So can the owner's Mac.
- **Board section.** The promotion needs a board section outside tables A–C. That is a local edit of the builder.

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3. Lane 2: the exponent, via PKM towers at m ≥ 4

3.1 Why this lane exists

The end-to-end cost model in `research/notes/index-calculus/RESEARCH_PKM_TOWER_ORACLE.md` §3.7 gives about r^{2/(m+1)} when the per-target cost is r^{o(1)}. So no oracle beats rho at m ≤ 3, and m = 4 is the first cell that could matter (derived there). The tower base is the prime-field base whose low-degree structure that note studies for a Gröbner solver. D1 notes the contrast: deg F_V = |V|, so the small-abscissa base gives a solver no low-degree structure.

3.2 `fall_degree_relaunch/`: relaunch the prime fall-degree study

Delivers.

- the relaunch card (`CARD_DRAFT.yaml`), with a proposed image, entry script and nine job templates under `proposed/`;

- draft Amendment 7, for t ≥ 12 (`AMENDMENT_7_DRAFT.md`);
- draft Amendment 8, the m = 4 extension (`AMENDMENT_8_DRAFT_m4.md`);
- an access request for a command-request kind (`ACCESS_REQUEST.md`), not performed;
- the Mac recovery steps (`MAC_RECOVERY.md`);
- 18 inconsistencies in the preregistration and launch files (README §5).

What it found.

- **Route (code reading).** `isolab.eks.request/v1` runs only ecbench specs. The only route that takes a command plus an image is `isolab_k8s_run`, which runs on the harness host's MCP.
- **Reproducibility check (measured).** The t ≤ 10 rows can be checked only at median level (3,4,4,4,4,5,5,5,5,6, from `research/prime_fall_degree_20261027/RESTART.md`), because the raw rows are on the owner's Mac.
- **t = 11 is not blind (measured).** The committed round-2 rows already gave D = 6 at t = 11 before Amendment 6 was registered (`research/pkm_tower_round2_20260925/runs/`). Only t = 12–14 test the C-step prediction, and t = 14 is the decisive cell.
- **Budget (derived).** Under the registered 1800 s F4 budget, t = 12–14 will run out: extrapolated F4 times are about 1.7 h, 17 h and 7 days. At t = 14, λ = 0.133, so about 12% of draws are decomposable and have no d_F. Amendment 7 proposes a two-cap bracket, larger resources, a rule for decomposable targets and a scorer patch.
- **m = 4 at N = 16 (derived).** The recorded F4 data give d_F = 7 at m = 4, N = 16, conditional on the G3 equivalence. Every engine asserts $p < 2^{32}$, so at the toy prime only about three tower levels above t = 4 are measurable. That cannot resolve one degree rise per four levels.
- **m = 4 at N = 20 (derived).** Uncapped F4 is projected at 240–380 GB, above the lab's 128 GiB pool cap.
- **Cryptographic size (derived, optimistic model).** At $p = 2^{256}$ and m = 4, a bounded d_F = 7 leaves about 25 bits to rho. A d_F that keeps growing like C-step leaves about 143 bits.

Decides. Whether the PKM-tower exponent lane stays open, under rules fixed before data in Amendment 8 (plan):

- **Close** if C-step holds at t = 12–14 and the m = 4 closure reads d_F ≥ 8 at N = 20.
- **Keep, as a mechanism question for m ≥ 6 only,** if d_F stays at 7 at m = 4.
- **Undecided at m = 4** for any other reading.

None of these outcomes is an S improvement or a leaderboard row (the study's `PREREGISTRATION.md` §8).

Blockers.

- **Submission route.** Submitting needs a session on the Iso Lab harness host, or the command-request change in `ACCESS_REQUEST.md`. Unblocked by the owner or the infrastructure owner.
- **The image.** It has never been built or smoke-tested. Unblocked by whoever holds ECR push rights.
- **The registered settings.** Under them, t ≥ 12 cannot be scored. Unblocked by the owner ruling on Amendment 7, plus a committed scorer patch.
- **The Mac-only raw data.** Unblocked by the owner, in person (`MAC_RECOVERY.md`).
- **Memory for m = 4 at N = 20.** It needs a 256–512 GB host, a disk-backed echelon, or an administrative decision on the pool cap. The cap should not be raised implicitly.

3.3 Phase 2 of `reference_arm/`

`pkm-semaev` with an msolve engine, on the `kummer-P22/P24` curves, would be the published tower arm. It waits on the msolve engine from §2.2. The derived F_p S_m-symmetrisation arm and an m = 4 chain probe wait on owner decision D5.

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4. Lane 3: constant engineering

4.1 `hashtable_prefetch/`: the hash-table floor of the relation trial

Delivers. A design (`README.md`), a patch (`patch.diff`, **uncompiled and untested**) and a stage split with its figure (`stage_split.py`, `.csv`, `.svg`).

What it found.

- **The lead's 137 ns (measured, open PR #2806, branch head `db02dd4f`, `research/ic_wide_ext_20261028/results.json`; not on `main`).** At 28 bits with 32 lanes, the guard insert costs 56.3 ns and the oracle probe 80.9 ns of a 318.1 ns trial. These are TSC cycles converted at 2.1 GHz, and they account for the 137 ns floor in `LEAD-ICP-prime-ic-constant-engineering`.

- **The real floor (derived).** The lane round also pays for a demand-read "touch" of both hash maps (30.6 ns at 28 bits and 47.9 ns at 32 bits, both at 32 lanes). That puts the real hash-table floor near 168 ns at 28 bits and 188 ns at 32 bits, about 53% of the trial.
- **Limits of the measurement.** The diagnosis cannot separate memory stalls from dispatch and timer cost: the cloud host has no PMU.

Design A (plan). Design A is an opt-in `mitm-prime-batch:m=2,layout=lines`, with three changes:

- the guard set and the pair table become 64-byte-line open-addressing tables;
- each key's home line is a pure function of the key;
- the existing touch becomes `prefetcht0`.

Expected gain (derived). Anywhere from under 5% to a relation-phase ratio of about 0.73–0.90×. S and the §5a band cannot move by construction, because records must stay identical.

Draft protocol (plan). One isolab-eks L2 session per size, at fp20 to fp32, with arms incumbent, A/A and candidate, plus walk-arm layout-only controls. It counts as an improvement only if the A/A-corrected relation-phase interval lies wholly below 1. The item closes if the interval spans 1 at both fp28 and fp32.

Decides. The next constant improvement on the generic track. Under R1 = Option C it counts and ratchets there, labelled generic, because it cannot beat rho.

Blockers.

- **Never compiled.** It needs a build host for `cargo fmt`, `clippy` and `test`.
- **Wrong base for `main`.** It is based on the open PRs #2806 and #2783, so it does not apply to `main`. Unblocked by merging them, or by a decision to stack on them.
- **No EKS identity here.** Running needs one, through the director/executor card flow.

4.2 The 20-bit factor-base row

This is the measured constant gain from §1. It ratchets the generic-track baseline once the second-host replay in §2.3 is identical. Two items in the study's own README come before any further claim on it: the replay, and the random-base control the study records as absent.

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5. Cross-cutting tracks

5.1 The lead atlas (crypto-autoresearcher `knowledge/leads/ecdlp/`)

The atlas track folded the whole `tools/lead_atlas.py --delta` into the atlas. After the review's edits, `--check` passes with 148 leads and 2092/2092 records mapped (as reported by the review). The track:

- added one new lead, `LEAD-ICP-pkm-tower-algebraic-oracle` (`open_blocked`);
- closed `LEAD-ICP-coppersmith-smallroot-semaev` as `closed_obstructed`. The closure is scoped to single-block convex lattices: with equal per-variable bounds a lattice reaches at most 0.3293 / 0.3536, against a demand of 8/9 (`research/smallroot_optimal_reach_20261026`);
- set the constant-engineering lead's best measured result to the 20-bit row;
- recorded the R1 scope notes.

`--check --strict-paths` still fails with 7 errors. They predate this review: they are paths from the unmerged PRs #2783, #2806 and #2807. Nothing is committed. The 49 lead files, `coverage.tsv` and `REVIEW_STATE.json` belong in one crypto-autoresearcher PR, together with the atlas report and graph refresh.

5.2 `review/`

The review checked about 35 cited numbers and paths against their sources, and all matched apart from wording and rounding slips, which it fixed. It added `DECISIONS.md` status notes to each track. It flagged the D2-versus-Option-C conflict but did not resolve it. It also corrected three stale atlas claims:

- the PKM lead pointed at the FGHR arm;
- the root-finding lead called itself the "default oracle" of Tables A and B;
- the constant-engineering lead said Table B was "unvalidated".

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6. Owner decisions still needed (deduplicated)

Already answered, so not re-asked here:

- the classification of root finding (R1 = C);

- whether the small-abscissa base counts (D1, tentative yes);
- the counter patch (D3);
- the reference-arm compute (D-EKS);
- whether a prefetch gain or the 20-bit gain counts: yes, on the generic track;
- the track of the Table B rows: generic.

id	decision	raised by	blocks
O1	~~Reconcile D2 with Option C.~~ **Resolved 2026-10-29:** D2 (owner) takes precedence. Ranking across tracks is allowed, and each track ratchets only its own baseline (`DECISIONS.md` R1). Remaining work: update rule C3 in `reference_arm/PROTOCOL.md` and `protocol.draft.json` at the freeze.	review, `reference_arm`, `ruling` §5.3	—
O2	**Confirm R1** by merging the PR that carries `DECISIONS.md` and the Option C wording for `AGENTS.md`.	`ruling`, atlas	the generic/algebraic relabel in SWEEPER, sweeper code and board
O3	**Details of the algebraic-track definition.** Does a lattice-window base count as algebraically defined? What share of grid-decided (`budget_exceeded`) systems still lets a `pkm-semaev` row count as algebraic? Confirm D1 at the freeze.	`ruling`, `reference_arm`	counting the first algebraic rows
O4	**D5:** are the F_p S_m-symmetrisation arm and the m = 4 chain probe in scope for phase 2?	`reference_arm`	phase 2
O5	**The fall-degree amendments.** Adopt, edit or reject Amendment 7 (two-cap bracket; `--budget 86400`; 56/112 Gi; decomposable draws excluded; scorer patch). Approve the corrected image and entry script, and set the per-job `seconds`. Adopt, edit or reject Amendment 8 (m = 4): widen `hky_fp_tower` to 64-bit primes, build an inconsistency certificate, or accept that the toy-prime m = 4 ladder cannot resolve the rate. Drop or condition the Stage B cell at m = 6, t = 6 (λ = 1.52).	`fall_degree_relaunch`	every t ≥ 12 row and every m = 4 row
O6	**Route for jobs that are not ecbench specs.** Option 0: the harness-host `isolab_k8s_run` only. Option 1/1b: an `isolab.eks.command/v1` request kind plus an ECR push route; this lets an authorized caller run any command in a lab-ECR image.	`fall_degree_relaunch`, `table_b_gate`, atlas	the fall-degree relaunch; the second-host replay
O7	**The Mac files.** Recover the Mac-only fall-degree files, or declare them lost. If lost, the t ≤ 10 check stays at median level.	`fall_degree_relaunch`, atlas	row-level reproduction check
O8	**The 20-bit second-host replay.** Authorise the replay of `ECBS1h685271327a8a` and choose its host: EKS `isolab_k8s_run`, or the owner's arm64 Mac. Replay only the holdout session, or the train-P20 selection sessions too?	`table_b_gate`, atlas	landing the 20-bit row
O9	**The Table B `pricing_ok` caveat.** P1: supersede the 2026-10-19 sweep line or render the per-row band (accounting only). P2: re-measure in one counted unit (new rows). P3: keep it permanently.	`table_b_gate`	removing the caveat
O10	**The prefetch run.** Authorise the L2 run, which D-EKS does not cover. Choose the incumbent oracle: `mitm-prime-batch` (the patch's target) or `mitm:negation_folded=1` (today's incumbent, which needs extra work). Stack on #2806/#2783, or wait? Make `layout=lines` the default if it wins? Add a cell on the ICV1 holdout curves?	`hashtable_prefetch`	the prefetch measurement
O11	**Atlas dispositions.** List the residual collector as its own labelled arm (generic track under C)? Confirm the scoped closure of the Coppersmith lead: its unequal-bound value 1/2 is numerical only.	atlas	atlas dispositions
O12	**Memory for m = 4 at N = 20.** Provide a large-memory host (256–512 GB, up for days), or rule on the 128 GiB pool cap.	`fall_degree_relaunch`, atlas	the decisive m = 4 cell

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7. Blockers and who unblocks them

blocker	affects	unblocked by
No cargo build: about 900 MB free disk in this container	reference-arm code, prefetch patch, ecbench builds	a session on a build host with several GB free
No pinned msolve build sound above 2^{16}	reference arm, PKM phase 2	an implementation session (pinned static build plus self-test)

blocker	affects	unblocked by
ecbench image and CI replay workflows lack msolve	running and replaying the reference arm	an implementation PR merged to `main`; the trusted publisher runs only from `main`
No AWS/EKS identity here	the reference-arm run, the second-host replay, the prefetch run	an executor with the lab identity (director → executor card flow)
Request route runs ecbench specs only	the fall-degree relaunch, the second-host replay	a harness-host session, or O6 Option 1
Raw fall-degree runs exist only on the owner's Mac	t ≤ 10 row-level check	the owner (O7)
m = 4, N = 20 needs more than the 128 GiB pool cap	the decisive m = 4 cell	the owner or the infrastructure owner (O12)
Prefetch patch is based on open PRs #2806 and #2783	applying and testing the patch	merging those PRs, or a decision to stack (O10)
No Conductor task attached in this container	every code edit outside this directory	the main session (`conductor check`)

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8. Ordered next steps

The order puts cheap, unblocking work first, and gates each measurement on the decision it needs.

1. **Main session: land the prep.**
   - Record this program in `docs/session-logs/2026-10-29.md`.
   - Under a Conductor task, open one crypto PR, ready for review with auto-merge, carrying this directory and `DECISIONS.md`.
   - Open one crypto-autoresearcher PR with the atlas edits and the atlas report and graph refresh.
   - Merging the crypto PR confirms R1 (O2).
2. **O1 is resolved (ranking across tracks is allowed; see `DECISIONS.md`).** Then the main session, under a Conductor task, does the relabel:
   - put the Option C wording in `AGENTS.md`;
   - update the `docs/ic/SWEEPER.md` tracks table;
   - add `Track::Generic` and retire `Track::Unclassified` in `src/bin/ic/sweep/space.rs`;
   - relabel the board, and fix the stale line 2380 in `scripts/build_ic_leaderboard.py`.

   This is an accounting change, not an improvement.
3. **Second-host replay of the 20-bit session (after O8, O6).**
   - Card: `table_b_gate/README.md` "The missing piece".
   - If 672/672 replays are identical, land the row as a generic-track section outside tables A–C, with `replay-plan.json` entries.
   - Add the random-base control to the next prime oracle study.
4. **Reference arm, implementation (no further decision needed; D-EKS covers the run).** On a build host:
   - build the pinned msolve and run its self-test;
   - implement `IMPLEMENTATION.md` §2;
   - run the unit tests, a 12-bit scratch session and `ecbench verify --replay`;
   - run the preflight, which measures runtime and RAM in place of the derived figures.
5. **Reference arm, run (after O3; O1 is resolved).**
   - Fill the FILL-AT-FREEZE fields, then freeze the protocol.
   - Check in `.resume/RESUME.md` (`scripts/checkpoint.sh init`).
   - Run through the experiment-director card and executor-mechanical on isolab-eks.
   - Land the report, graphs, PDF, `VALIDATION.json` and the `benchmarks/` entry.

   This is the first algebraic-track row.
6. **Fall degree (after O5, O6; O7 in parallel).**
   - Commit Amendment 7 and the scorer patch.
   - Build and smoke-test the image, check in the resume checkpoint, and freeze the card.
   - Submit jobs A and C, then B, then the t = 12 brackets, then t = 13 and t = 14 one at a time.

- Score t = 12–14 against C-step.

7. **Prefetch (after O10, and #2783/#2806 merged or a stacking decision).**
   - Compile and test the patch.
   - Confirm that #2806's ten certified sessions replay bit-identically (1,176 replays, plan).
   - Freeze the protocol and run the L2 sessions.
   - Apply the pre-declared rule.
8. **m = 4 extension (after Amendment 7's reading, O5 Amendment 8, O12).** Register Amendment 8 before any m = 4 row at t ≥ 4. Then run the t = 4 closure to c = 7 first.
9. **Phase 2 and housekeeping.**
   - Add the PKM msolve arm, plus the O4 items if approved.
   - Add a Table B entry to `docs/ic/validation/reference-currency.json`.
   - Extend replay-plan coverage to the `ic boundary` round-5 ledger.
   - Fix the atlas `--strict-paths` errors once #2783/#2806/#2807 merge.
   - Resolve the Table B `pricing_ok` caveat per O9.

---

9. Files

path	content
`README.md`	this program overview
`program.svg`	program diagram (hand-written SVG; light and dark via `prefers-color-scheme`; system fonts only)
`REPORT.pdf`	rendering of this README and the diagram (see below)
`build_pdf.py`	the script that rendered `REPORT.pdf`
`DECISIONS.md`	owner decisions, 2026-10-29
`ruling/`	rootfind classification memo and `classification.json`
`reference_arm/`	published algebraic reference: recommendation, draft protocol, implementation spec
`table_b_gate/`	validation-gate dossier for Table B and the 20-bit row
`fall_degree_relaunch/`	relaunch card, Amendments 7 and 8 (drafts), access request, Mac recovery, proposed image and jobs
`hashtable_prefetch/`	hash-table floor design, uncompiled patch, stage split
`review/`	cross-track review

About the PDF. pandoc 3.1.3 is installed, but none of its PDF engines are: `pandoc README.md -o REPORT.pdf` fails with "pdflatex not found", and `--pdf-engine=` wkhtmltopdf, weasyprint, typst, pdfroff and context are absent too. So `REPORT.pdf` was **not** built by pandoc. It was rendered by `build_pdf.py` with reportlab 5.0.1, the only PDF library present. That script reads this README's text and `program.svg`, and draws the diagram itself. Its typesetting is plain: tables are text grids, and only the light colour scheme is used. For the research-visuals PDF, re-render with `pandoc README.md -o REPORT.pdf --pdf-engine=typst` on a host that has typst, when this directory lands.